Fraud Transactions Analysis Report (2025)

Prepared 25 October 2026 · Brian Mbae Mwarania

Executive summary

Fraud affected 1.2% of the 60,000 transactions in 2025 (720 cases, KES 2,665,780), and it is concentrated in a few clear segments rather than spread evenly.

- **Channel:** Card Online runs at 3.68%, about three times the average. M-Pesa and Card POS sit near 0.6%.
- **Time:** between 00:00 and 04:59 the fraud rate is 6.7% to 9.2% per hour, against roughly 0.3% to 1.1% for most of the day.
- **Geography:** the US (17.2%), UAE (16.4%) and UK (15.0%) are more than ten times the average. Kenya is the only country below it, at 0.93%.
- **Merchant category:** Electronics (6.1%), Betting (3.4%), Travel (2.7%) and Transfers (1.4%) are above average.

These are single-variable views. The next step is to test how the factors overlap, for example whether night-time card-online spend abroad drives most of the loss.

Dataset and data quality

The dataset holds 60,000 Kenyan-shilling transactions from 1 January 2025 (00:01:34) to 31 December 2025 (23:47:11), with no missing values and no duplicate rows or transaction IDs.

Column	Type	Description
transaction_id	text	Unique ID (TX0000001 onwards)
customer_id	integer	Customer identifier
timestamp	datetime	Converted from text and sorted chronologically
amount_kes	decimal	Transaction value in KES
merchant_category	text	10 categories (Utilities, Airtime, Fuel, Restaurants, Groceries, Electronics, Health, Betting, Travel, Transfers)
channel	text	M-Pesa, Card POS, Card Online, Bank Transfer, ATM
country	text	KE, US, AE, UG, TZ, GB
new_device	0/1	Whether a new device was used
is_fraud	0/1	Fraud label (target)

An `hour` field (0 to 23) was derived from the timestamp for the time-of-day analysis.

Overall fraud rate

Of 60,000 transactions, 720 were fraudulent, an overall fraud rate of 1.2%. This figure is the baseline that every segment below is compared against.

Fraud by channel

Card Online has the highest fraud rate at 3.68%, roughly three times the 1.2% average, and it is the only channel clearly above it.

Channel	Transactions	Fraud cases	Fraud rate
Card Online	9,267	341	3.68%
ATM	6,004	69	1.15%
Bank Transfer	9,091	96	1.06%
Card POS	11,924	72	0.60%
M-Pesa	23,714	142	0.60%

Fraud cases are derived from each channel's rate and volume. M-Pesa is the busiest channel but one of the safest per transaction. Card Online accounts for about 47% of all fraud cases from 15% of transactions.

Fraud by hour of day

Fraud is overwhelmingly a night-time problem: every hour from 00:00 to 04:59 has a fraud rate between 6.7% and 9.2%, five to eight times the 1.2% average.

Hour starting	Fraud rate
03:00	9.22%
04:00	8.87%
02:00	7.61%
01:00	7.22%
00:00	6.65%
05:00	1.90%

The rate drops sharply after 05:00. For the remaining hours it ranges from 0.34% (13:00) to 1.12% (06:00), so most of the day sits at or below the average. The hours with the highest rates are not necessarily the busiest, so counts by hour are worth checking before setting rules.

Fraud by country

Transactions made from outside East Africa carry a fraud rate above 15%, while Kenya, the home market, is the only country below the 1.2% average.

Country	Fraud rate
United States (US)	17.15%
United Arab Emirates (AE)	16.36%
United Kingdom (GB)	14.97%
Uganda (UG)	3.35%
Tanzania (TZ)	3.24%
Kenya (KE)	0.93%

Uganda and Tanzania are about 2.7 times the average, far below the three overseas markets but still elevated. Transaction counts per country were not computed, so the share of total fraud coming from each market is not yet known.

Fraud by merchant category

Electronics is the riskiest category at 6.10%, about five times the average, followed by Betting and Travel; four of the ten categories exceed 1.2%.

Merchant category	Fraud rate	Versus 1.2% average
Electronics	6.10%	Above
Betting	3.41%	Above
Travel	2.74%	Above
Transfers	1.39%	Above
Restaurants	0.87%	Below
Health	0.77%	Below
Airtime	0.69%	Below
Fuel	0.57%	Below
Utilities	0.27%	Below
Groceries	0.26%	Below

Everyday spending (Groceries, Utilities, Fuel, Airtime) is the safest. The risky categories are high-value or easily resold goods and services.

Financial impact

The 720 fraudulent transactions total KES 2,665,780, an average of about KES 3,702 per fraud case. The notebook does not yet report total transaction value, so fraud as a share of value is still to be calculated.

Recommendations

The findings point to four controls that target the highest-risk segments without adding friction to the safe majority. Each should be tested against historical data before rollout.

1. **Night-time step-up checks:** require extra authentication, or a short hold, for transactions between 00:00 and 05:00.
2. **Card Online protection:** apply 3-D Secure or tighter velocity limits to online card payments, which carry 3.68% fraud.
3. **Cross-border rules:** flag or challenge transactions from the US, UAE and UK, where rates exceed 15%, particularly for customers who normally transact in Kenya.
4. **High-risk categories:** add scoring or limits on Electronics, Betting and Travel spend, with Transfers monitored.

Limitations and next steps

This analysis looks at one factor at a time, so overlaps between factors are untested and the strongest rule may be a combination.

- Analyse the `new_device` flag and `amount_kes` distribution (fraud versus legitimate) against fraud.
- Cross-tabulate channel, country and hour to find the combinations that explain most fraud.
- Add transaction counts and fraud counts alongside rates for hours, countries and categories, so small segments are not over-read.
- Look at repeat customers: some IDs appear more than once among early fraud cases (for example customer 4052).

- Build a classification model (for example logistic regression or gradient boosting) and evaluate it with precision and recall, since fraud is only 1.2% of the data.